

AQI Feature Store Architecture

Purpose

The feature store provides a consistent interface between the AQI feature pipeline, model training pipeline, and online prediction service.

Architecture

External Weather/AQI APIs

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Feature Collection

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Validation

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Feature Engineering

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Historical Features Latest Features

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Parquet Offline Store SQLite Online Store

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Feast Registry

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ML Training Prediction API

Offline Store

Historical engineered features are stored in Parquet files and used for training and historical feature retrieval.

Online Store

SQLite is used as the local online feature store for the internship implementation.

Leakage Prevention

The following forecast labels are not included in the online Feature View:

- target_aqi_24h
- target_aqi_48h
- target_aqi_72h

These values represent future observations and are used only during supervised model training.

Forecast Horizons

The system predicts:

- AQI +24 hours
- AQI +48 hours
- AQI +72 hours